

DEEP LEARNING

MCTA 4363

SECTION 1

ASSESSMENT 2 REPORT

Gemma 4 E4B Compression &
Optimization (Track 3)

VISCA BARCA

SEM 2 2025/2026

1. Executive Summary

This report details the development and optimization of a vision-language solution using the Gemma 4 E4B (Image-to-Text) model. The primary objective was to maximize energy efficiency and minimize memory footprint while maintaining a high quality of inference. Through iterative testing of quantization, Mixture-of-Experts (MoE) routing, and token selection, we are developing a pipeline capable of running on standard 16GB VRAM hardware with a significantly reduced carbon footprint.

2. Experimental Iterations & Comparisons

This section documents the evolutionary steps of model development.

Experiment 000: Baseline Model

- **Objective:** Establish empirical baseline metrics for Gemma 3n E4B on a stratified VQAv2 sample before applying any compression techniques.
- **Implementation:** Loaded google/gemma-3n-E4B-it on Google Colab T4 in bfloat16. Fed 50 stratified VQAv2 samples (15 yes/no, 15 counting, 20 open-ended). Measured TTFT via Python `time.perf_counter()` and energy via CodeCarbon.
- **Result:** VQA Accuracy 36%, Avg TTFT 12.44s, CO₂ per sample 0.124g.
- **Decision:** Pivoted to Gemma 4 E4B as the correct competition model. Carried forward the CodeCarbon + VQAv2 methodology.

Experiment 001: Raw Baseline (Updated)

- **Objective:** Establish a performance baseline using the `google/gemma-4-31B-it` model on standard hardware.
- **Implementation:** Attempted a direct load using the `transformers` pipeline for image-text-to-text tasks.
- **Failure Analysis:** The 31B model requires approximately 62GB of VRAM in bf16, which exceeded the 14.56 GB free capacity on the T4 GPU, leading to an immediate `OutOfMemoryError`.
- **Pivot:** Identified the correct competition model as `gemma-4-E4B-it`, which utilizes a Mixture-of-Experts (MoE) architecture with 4B effective parameters.

Experiment 002: NF4 4-bit Quantization (The Round 1 Submission target)

- **Strategy:** Implement 4-bit Normal Float (NF4) quantization to fit the model within the 16GB VRAM constraint.
- **Implementation Details:**
 - Used `BitsAndBytesConfig` with `load_in_4bit=True` and `bnb_4bit_quant_type='nf4'`.
 - Enabled `bnb_4bit_use_double_quant=True`, which quantizes the quantization constants to save an additional ~ 0.4 bits/parameter.
 - Utilized `device_map='auto'` for intelligent memory offloading.
- **Current Status:** Pipeline built and code validated on Colab T4. Model confirmed loading successfully. Quantitative VQA accuracy and CO₂ benchmarking pending — scheduled on RTX 4060 before May 16 Round 1 submission window.
- **Expected outcome:** $\sim 52\%$ VRAM reduction based on NF4 literature. Accuracy target $\geq 80\%$ of baseline per competition threshold.

3. Methodology Comparison Table

Experiment	Technique	VRAM after (GB)	CO2/Sample	VQA Accuracy Status	Average TTFT
000	Baseline	10GB	0.124g	36%	12.44s
001	Raw 31B Pipeline	Out Of Memory	N/A	N/A	N/A
002	NF4 4-bit (E4B)	4.8 GB	TBD	In Process	TBD

4. Technical Implementation Details

The current pipeline utilizes a multi-layered compression stack:

- **Quantization:** Normal Float 4-bit (NF4) with Double Quantization via BitsAndBytesConfig. Quantizing the quantization constants saves an additional ~0.4 bits/parameter on top of standard 4-bit compression.
- **Precision:** bnb_4bit_compute_dtype set to torch.bfloat16 — weights stored in 4-bit, computations in bfloat16 for numerical stability.
- **Energy Monitoring:** CodeCarbon integrated to track real-time GPU and CPU emissions via NVML, matching the competition's official evaluation methodology exactly.
- **Serving:** vllm_config.yaml prepared for vLLM 0.17.1 serving on NVIDIA L4 (16GB), as required by the competition evaluation pipeline. Full quantitative benchmarking of VQA accuracy, TTFT, and CO₂/sample scheduled on RTX 4060 before the May 16 Round 1 submission window.

5. Github

<https://github.com/Syaz03/MCTA4363-Resilient-AI>

6. Interim Submission

Interim submission will be done around 19-20 May 2026 (Via technical briefing done on 27th April 2026)



Sustainable AI Coalition <contact@sustainableai.coalition.org>
to Resilientchallenge2026 ▾

12:24 AM (7 hours ago) ☆ 😊 ↩ ⋮

Dear participant,

👉 We're thrilled to share that the submission deadline for Round 1 for the audio-to-text and text-to-text categories **has been extended to May 9, 2026**. This gives you additional time to submit your models, don't miss this opportunity!

Why submit now?

- **Even incomplete models are welcome.** The mid-challenge leaderboard **is not final and does not reflect the ultimate evaluation**
- **Feedback opportunity:** Early submissions will help you assess where your model currently stands in the competition!

You can find all details for a seamless submission in our previous emails. **Please ensure you complete all stages to guarantee a successful submission.** Be aware that the submission process can be slightly different for the text-to-text and the audio-to-text categories. Thus, **make sure to check the proper email if you are registered in both categories** (you should have received one email per category, except for the image-to-text category: details will be shared with you later).

⚠ Some of you have reached out as they were unable to grant us access to their model on Hugging Face. We are now aware of this issue and we've found an alternative to help you submit your model. **Thus, if you encounter any issue, please contact the evaluation team at resilientchallenge2026@peren.gouv.fr**

In your email please indicate: **the name of your team and the URI of the model**

With best regards,

The Resilient AI Challenge Team

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The Sustainable AI Coalition Coordination Team